

# Appendix X

## Detailed Thematic Analysis of the Perceived Strengths of LitTraceAI

### X.0.1 Transparency and Citation Integrity

Transparency Features captures how experts valued LitTraceAI’s ability to make AI-generated content inspectable, traceable, and interpretable. Experts frequently associated these features with greater confidence in the generated literature review section because they could verify how claims were constructed from the set of papers.

19 out of 27 experts appreciated the presence of contextualised explanations showing how generated sentences were produced. Expert 8 described the provenance functionality as *“extremely powerful ... because it also shows me how everything was generated.”* Similarly, Expert 17 explained that *“the generated information is justified, especially for people like me, who need to see that something is true.”*

Other experts similarly associated explanation features with transparency and trust. Expert 3 stated that *“the provenance ... definitely makes me trust the tool”*, adding that showing *“how the tool generated the sentence and its source text fosters transparency.”* Likewise, Expert 19 described provenance as *“the most powerful aspect of the tool”* because *“it makes*

*the creation process transparent and shows me the source.”* The expert further explained that this was particularly important *“for people like me who are very distrustful.”*

These perspectives indicate that experts perceived transparency as supporting trust through visibility into the generation process.

One expert contrasted this capability of LitTraceAI with experiences using other AI tools. Expert 16 explained that a previous interaction with ChatGPT produced a statement that was incorrect and later justified through what the expert described as *“fabricated reasoning that did not align with the document.”* In contrast, the expert highlighted the value of LitTraceAI because it provides *“contextualised and fine grained justifications.”* He further stated:

*What I find particularly valuable here, by contrast, is your [LitTraceAI] approach of providing contextualised and fine grained justifications. This adds real value and goes beyond what even top tier large language models typically offer.*

This observation indicates that experts may perceive explicit explanations of how generated sentences are constructed from source material as a distinctive feature that is not commonly available in other AI tools. Expert 11 similarly highlighted the uniqueness of combining source traceability with generation explanations, stating that *“the fact that it [LitTraceAI] tells me which are the source sentences it used to create the sentence and also indicates how it [LitTraceAI] created the sentence gives me confidence and it is something unique.”*

18 out of 27 experts highlighted the importance of directly accessing the original source sentences. Expert 4 referred to *“the ability to search for the source of what is being written directly in the paper”*, while Expert 13 stated, *“Here I can see where the tool got the original texts.”* Expert 6 further explained:

*I see value in provenance because I usually waste a lot of time verifying information from the original paper. Instead, with this provenance feature, if the tool*

*[LitTraceAI] tells me it [generated sentence] was taken from here [source sentence], then I can read the exact sentence and determine whether what the AI rewrote actually preserved the meaning and maintains full coherence with the paper.*

Expert 27 similarly emphasised the importance of provenance as a mechanism for verification and understanding, stating: *“Provenance ... I need it for any kind of generative AI tool. I need the ability to check things. I need some trust in the system. I need to be able to develop an understanding of how it works.”*

These reflections indicate that experts used the original source sentences as a verification mechanism to assess whether the generated content was grounded in the original papers and whether the rewriting remained faithful to the source material.

Although experts generally valued the explanation features, some comments also indicated that provenance still required manual review. Expert 26 explained: *“The tool [LitTraceAI] explains to me how it generated the text and where it sourced the information, although it needs to be reviewed. That is to say, I would have to dedicate more time to go through the explanations one by one to check whether they are correct, but the ones I saw are correct.”* This observation suggests that experts still considered human validation necessary when assessing the correctness of generated content, despite valuing the transparency mechanisms provided by LitTraceAI. This perception aligns with an **AI-in-the-Loop** approach.

8 out of 27 experts also valued the categorisation of sentences according to their rhetorical role. Expert 15 stated that *“if I wanted to group the gaps or limitations of the studies, this tool makes it easier for me.”* Likewise, Expert 12 explained that *“the classification of sentences gives me a guideline of the order and structure of the text as well as the gaps in the literature.”* Expert 7 similarly noted that *“the classification of sentences is key because it gives you an idea of the gaps in the literature.”*

Expert 27 highlighted additionally that sentence classification reflected rhetorical structures already embedded within scientific writing:

*The sentence classification, for example, knowing that this is what's going on underneath, makes sense. These rhetorical aspects of papers are inherent to them. People may quibble about the exact details, but this is what's real about papers. The tool is using the structure that's already there and making it evident, reorganising it, and so on.*

This perspective indicates that experts may perceive sentence classification as supporting not only organisation, but also the interpretation of rhetorical and analytical relationships within the literature.

5 out of 27 experts appreciated indicators showing the strength of citation support. Expert 21 noted that provenance “*shows you the type of support for the citation*”, while Expert 20 stated that it “*gives me the confidence level of the generated sentence with the citation accuracy.*” In LitTraceAI, citation support was represented through different accuracy levels, including *Full Support*, *Partial Support*, and *No Support*. These reflections suggest that some experts valued the possibility of inspecting how strongly a generated sentence was supported by the cited material. Rather than treating citations as uniformly valid, experts appeared to appreciate distinctions regarding the degree of support provided between the generated sentence and the source papers.

One expert additionally highlighted the visibility of transformation processes:

*You also show the transformation edits, which is great. And as you mentioned, you have this taxonomy that clearly explains what kinds of operations your tool [LitTraceAI] performs to produce the resulting sentence (Expert 16).*

Although this feature was mentioned by only 1 out of 27 experts, it appears analytically relevant because, as discussed in Section ??, plagiarism is a latent concern in the automatic generation of literature reviews. This perspective indicates that making transformation operations visible may support users in evaluating whether the generated text reformulates

the original material appropriately, or whether the output risks reproducing content too closely from the source papers.

Citation Accuracy refers to how experts perceived LitTraceAI as reliably connecting generated sentences to authentic references from the provided papers. Experts valued the tool’s ability to avoid fabricated citations, particularly in comparison to other generative AI tools. Reference fidelity was identified by all experts contributing to this aspect (3 out of 3 experts). Although this aspect appeared with a relatively low frequency, experts considered it particularly relevant because rigorous referencing was perceived as fundamental to academic writing.

Expert 2 noted that LitTraceAI *“makes good use of citations and does not make them up like ChatGPT.”* This observation suggests that citation authenticity contributed to perceptions of trustworthiness in the generated literature review.

Likewise, Expert 21 associated reliable referencing with academic rigour:

*Rigour in the use of references. It does not fabricate references like ChatGPT does. I think it does a good job on this topic, which is fundamental in academic writing.*

Similarly, Expert 22 perceived the citation process as more reliable than in comparable AI tools, stating that *“the citation process works well; in other tools, this tends to fail quite often.”* The expert highlighted further that LitTraceAI *“writes by citing”*, which appeared to support a coherent flow of ideas across the generated text. This perspective may indicate that experts not only valued citation authenticity, but also appreciated how citations contributed to the structural continuity of the literature review.

These perceptions may be associated with the design of LitTraceAI described in Section 4.2.1, where the generated content is grounded exclusively in the references contained within the provided set of papers. This mechanism may help reduce citation hallucinations commonly associated with other AI tools, as highlighted by the experts.

Taken together, these perspectives indicate that experts viewed accurate citation attribution as an important characteristic of credible AI-assisted academic writing.

## **X.0.2 Scholarly Synthesis and Academic Organisation**

Logical Structure refers to how experts perceived the generated literature review as being coherently organised through thematic grouping, logical sequencing, and the use of structured headings. The findings indicate that experts valued not only the presence of organisational features, but also their role in making the literature easier to interpret and navigate.

Thematic heading generation was mentioned by 11 out of 17 experts. Several experts described how LitTraceAI divided the literature into meaningful conceptual sections that helped structure the discussion of prior research. Expert 12 noted:

*It focuses on the four most important topics. This classification by themes gives a mapping of the subject matter.*

This reflection may suggest that thematic grouping may support experts in developing an overview of the field. Likewise, Expert 13 described the process as “*grouping [sentences] into themes and summarising them into a logical structure,*” indicating that experts associated thematic segmentation with coherent synthesis.

Expert 15 explicitly highlighted this capability as something not commonly found in other AI tools:

*The division of themes. This is something the tools I have used do not do, and it does it very well. I am truly surprised.*

Additional experts reinforced the value of thematic clustering as a mechanism for structuring literature reviews according to recognised academic conventions. Expert 16 explained that “*the clustering of the topics is a plus in this tool [LitTraceAI] because it categorises topics with a heading, which is what we do in a related work section.*” Similarly, Expert 17

stated that *“the grouping of topics”* resembled *“the headings you place in the state of the art section.”* Together, these reflections may indicate that experts perceived thematic heading generation as a distinctive feature of the tool that supported the academic structuring of the generated content.

Beyond organisation, one expert suggested that thematic clustering could also stimulate further scholarly exploration. Expert 11 remarked:

*The clustering of topics is very good and gives me ideas for aspects that I can include in the future, or other topics that can be explored that I truly did not know could be related.*

This reflection may represent another example of selective acceptance. In Section 7.1.2, Expert 4 acknowledged that some generated headings were defensible based on the reviewed papers, but did not use them because they did not align with the intended focus of the literature review. In contrast, Expert 11 appeared more willing to accept additional thematic groupings because they suggested potentially relevant research topics and conceptual connections that could be incorporated into future work.

Experts also discussed the overall sequencing and rhetorical flow of the generated content (6 out of 17 experts). One expert explained that *“the weaving of ideas when presenting the content is highly synthesised”* and that the structure followed *“introduction, context, and research gap.”* Similarly, Expert 18 observed that the synthesis resembled *“the structure of a theoretical framework in a paper”* because *“it moves from the general to the particular...”*

These positive perceptions may indicate alignment with the design of LitTraceAI described in Section ??, where the literature review is organised through the classification of sentences from the set of papers into introduction, contextualisation, and research gap components following a progression from the general to the specific. This interpretation is also reflected in Expert 23’s statement that the tool generated *“a good logical structure”* that *“is easy to read.”*

The findings from the current analysis appear to complement those presented in Section 7.1.2. While Section 7.1.2 emphasised experts' evaluations of representativeness through thematic coverage, detail, and coherence between ideas, the current analysis indicates that experts also valued the organisational structuring of these themes through headings and clustering. Therefore, these findings may suggest that experts perceived representativeness not only in terms of content accuracy, but also through the coherent academic organisation of the literature.

Critical Analysis reflects how LitTraceAI supported experts in identifying relationships, similarities, and differences across studies, while also helping them recognise research gaps and limitations within the literature. Five out of seven experts referred to the tool's ability to establish relationships across studies, whereas two out of seven experts highlighted its support in identifying research gaps and limitations.

Several experts valued the comparative analysis generated by LitTraceAI. Expert 13 explained that *"the depth of analysis and reflection"* and *"the way it compares and contrasts is something very similar to what I would do."* In a similar way, Expert 1 highlighted *"the comparative analysis made between the authors of the papers"*, indicating that experts appreciated how LitTraceAI explicitly connected findings and perspectives across studies. These observations complement the findings in Section 6.2, where LitTraceAI demonstrated stronger analytical depth consistently through comparison between studies. This is particularly important because critical analysis is a fundamental characteristic of literature review, requiring not only the description of studies but also the integration and evaluation of relationships across the literature.

Experts also perceived value in the identification of research gaps and limitations. Expert 2 stated that *"the identification of gaps and limitations gives me an idea of where to start"*, suggesting that the generated output may support the early stages of research exploration. Likewise, Expert 19 explained that *"I can identify the research gaps through the classification of the sentences."* These observations also complement the findings in Section 6.2, where

LitTraceAI demonstrated stronger performance in identifying research gaps. Identifying gaps in prior research is a critical component of literature review, as researchers use these gaps to position their work and construct the narrative for motivating new research contributions.

### **X.0.3 Coverage**

The theme of Coverage reflects how experts perceived LitTraceAI as generating a broad and representative synthesis of the source papers while remaining closely grounded in the provided literature. Experts particularly valued the balance between comprehensive integration of the literature and fidelity to the selected corpus.

Regarding LitTraceAI remaining focused on the source papers (2 out of 5 experts), experts frequently contrasted the tool with general-purpose AI systems. Expert 22 explained:

*The tool [LitTraceAI] focuses exclusively on all the text content that it is fed. It is not like ChatGPT, which searches for information that is not relevant to the topic.*

Similarly, Expert 9 stated that LitTraceAI “*does not fabricate [papers] or include irrelevant ones, unlike ChatGPT, Perplexity AI, or SciSpace.*” These perspectives indicate that experts valued the fact that the tool generated the literature review from a predefined set of source papers that had already been selected by the researcher as relevant to the topic. In contrast, other AI tools were perceived as needing to search for relevant papers independently, which may increase the likelihood of introducing irrelevant information. This appears to be a distinctive characteristic of the tool, as experts perceived the generated content as remaining grounded in the selected literature. The reflections presented in this theme also reinforce the findings from Section 6.2, particularly regarding the perceived fidelity of LitTraceAI to the selected source papers.

Experts also emphasised the breadth and integrative capacity of the generated synthesis (2 out of 5 experts). Expert 22 described LitTraceAI as capable of connecting information across a large number of papers:

*In fact, I see something very positive here: you can upload several papers, even dozens of them, and the tool [LitTraceAI] manages to connect the information to tell you a story. That is something I have not been able to achieve with Copilot; I load three papers and the poor tool gets lost.*

This perspective may indicate that experts perceived the tool as capable of coherently integrating information across multiple source papers, even when processing a substantial body of literature. This observation also aligns with the evaluation of LitTraceAI in Section 6.2, where experts highlighted its ability to connect and synthesise information across studies without introducing irrelevant material.

Expert 27 also emphasised the value of the tool’s extensive coverage when writing a literature review:

*And the coverage is great. That’s something that’s really hard when you’re writing a literature review, because you have to read and read in order to digest everything. So the coverage is a huge strength.*

This observation may indicate that experts perceived LitTraceAI as supporting the demanding process of reviewing and digesting large amounts of literature when writing a literature review.

In follow up conversations, some experts further reflected on potential challenges associated with summarisation and corpus selection. These reflections were relevant to the analysis because they provided additional insight into how experts interpreted the strengths and boundaries of LitTraceAI beyond the initial evaluation session.

Drawing on his teaching experience with student-generated summaries, Expert 24 noted:

*Students tend to study because they are short on time, relying on summaries of the readings one assigns. But then, sometimes I tell them, well, be careful. Who made the summary? Because whoever makes a summary can include some things and leave out others: whatever caught their attention or seemed interesting to them. Anything, really. The same thing happens now with Artificial Intelligence: you have to be careful with the summary it produces, because it might leave out precisely the best argument.*

Rather than describing a direct limitation observed in LitTraceAI, this reflection may indicate a broader concern regarding the interpretive nature of summarisation. Expert 24's perspective suggests that, even when the tool produces coherent syntheses, experts may still consider it important to critically review whether key arguments from the source papers have been sufficiently represented. This view aligns with the **AI-in-the-Loop** perspective.

Similarly, in a follow up conversation, Expert 26 reflected on how the breadth and representativeness of the selected source papers may influence the comprehensiveness of the generated synthesis:

*I also think it is because the tool [LitTraceAI] does not have enough information. The topic has many perspectives and thousands of related variables. It is like, being human, you have a very incomplete map: you know some personality traits, fine, and suddenly you jump to organizational support. Sure, it may be that in the four papers the AI [LitTraceAI] analysed, that was discussed, but that does not mean it reflects the whole reality.*

This reflection was not necessarily presented as a limitation of LitTraceAI itself, but rather as a possible scenario in which a small or unbalanced set of source papers may reduce the breadth of the generated literature review. In this case, Expert 26 was referring to Unethical Pro-Organisational Behaviour, which is a complex topic within organisational behaviour involving multiple dimensions and related variables. This reflection aligns with

findings from Section 7.1.2, where experts distinguished between the quality of the synthesis produced by LitTraceAI and the representativeness of the literature provided as input. Similar to Expert 2 in Section 7.1.2, Expert 26 suggested that a broader or more diverse set of source papers would likely require a more comprehensive synthesis.

Finally, experts indicated that the generated output incorporated the principal concepts successfully from the source papers (1 out of 5 experts). As Expert 21 explained, *“The tool [LitTraceAI] captures the main ideas. From what I observed, it takes the texts and interprets them appropriately.”* This perspective corroborates findings reported in Section 7.1.2, where experts in the Excellent and Good categories highlighted the ability of LitTraceAI to preserve foundational themes and maintain strong thematic coverage across the analysed literature.

## **X.0.4 Usability and Accessibility**

Usability and Accessibility reflects how experts perceived LitTraceAI as both easy to interact with and visually accessible. Ease of interaction was identified by 1 out of 2 experts. Expert 5 described the tool as *“easy, friendly, and accessible”* and explained that *“it does not take long, I click and the information loads immediately!”* This observation suggests that the interface supported efficient interaction through minimal effort and straightforward navigation.

Accessibility and perceptibility were also emphasised by 1 out of 2 experts. Expert 5 evaluated the interface using the WAVE accessibility tool and stated:

*The tool [LitTraceAI] allows you to quickly review the usability side of things and also how it is structured. On top of that, I checked it with the WAVE accessibility tool, and it did not throw any contrast errors or other issues! This means that, at least in terms of accessibility principles, it is meeting the perceivable criterion, which is one of the most important ones for users.*

The expert explained further that readability depends on sufficient distinction between background and text colours, noting that *“a white background with black letters offers a perfect contrast.”* The expert stated additionally that *“if you zoom in from your browser, you will be able to see the text much better, especially if you have some kind of vision impairment”*, indicating that the interface remained readable and accessible under browser zoom conditions.

These findings corroborate the observations presented in Section ?? regarding usability and navigational simplicity. In both sets of findings, Expert 5 emphasised that LitTraceAI reduced interaction complexity by allowing users to access generated content quickly and with minimal navigation effort. This contrasts with other AI-based literature summarisation tools such as SciSpace, Elicit, and Consensus, which were perceived as overloaded with options and difficult navigation flows. The present findings extend the discussion beyond usability by indicating accessibility-related strengths, particularly regarding visual contrast and readability during browser zooming. Although the frequency of these observations is low, they remain relevant because they originate from an expert specialised in usability and accessibility.

As discussed in Section 2.2.6, usability and accessibility dimensions within AI-supported literature review generation remain relatively understudied. These findings may indicate that interface simplicity, perceptibility, and accessible visual presentation could represent relevant factors influencing how researchers interact with AI-generated literature review tools. While no broader conclusions can be drawn from a single expert perspective, the observations suggest that accessibility and usability warrant further exploration.